Bangladesh Army University of Science and Technology

Department of Computer Science and Engineering

Final Examination, Winter 2023

Course Code: CSE 3101 (Syllabus 2015) **Level-3 Term-I**

Time: 03 (Three) hours **Full Marks: 180**

Course Title: Database Management Systems

N.B.
- The questions are of equal value.
- Figures in the margin indicate full marks allotted to each question.
- Symbols and abbreviations bear their standard meaning.
- Use separate answer script for each PART.
- The corresponding course learning outcomes (CLOs) are given in the right most column.

PART- A (Marks: 90)

(Answer any three questions including Q. No. 1)

			Marks	CLOs
1.	a)	Suppose BAUST needs to build an automated system. The authority has hired you to design the required database. Due to this reason, you have visited the organization and conducted your own analysis over existing system. Design a database based on your findings. You must draw an ER diagram and reduce it into relational schemas.	15	CLO3
	b)	Briefly describe the ACID properties of a transaction.	10	CLO1
	c)	Define DBMS. Write some applications of DBMS.	05	CLO1
2.	a)	Consider the precedence graph shown in Figure 2(a) which consists of five transactions A, B, C, D and E:	10	

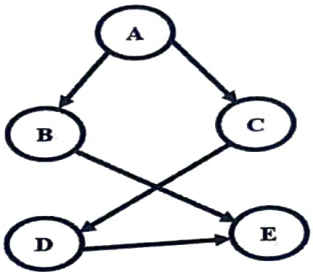


Figure 2(a): Precedence graph of a schedule

Determine whether the schedule is conflict serializable or not. Also, briefly explain the reason. If it is conflict serializable, then determine all the possible serializability orders using topological sort.

	b)	Justify the following statement – "*Varchar* is more memory efficient than *char* datatype".	05
	c)	*Products(pid: integer, name: string, cat: integer)* *Categories(cid: integer, name: string)* *Sale(sid: integer, pid: integer →Ref(Products: pid), cid: integer →Ref(Categories: cid), sdate: date, qty: integer, price: float)* For the schemas above, write down the relational algebraic expressions for the following:	
		i) Find the name of all products.	02
		ii) Select tuples from Sale where price is greater than 5000.	03
		iii) Find the names of the products that were sold before 5 May 2023.	05
		iv) Find day wise average price of each sale.	05
3.	a)	Define cryptography. Briefly describe the components of cryptosystem with proper block diagram.	10

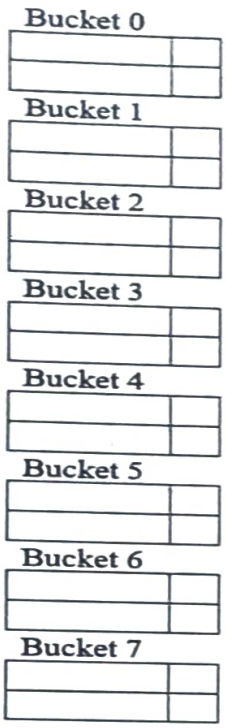
b) Consider, a database table has a total of 2500 records which are organized in sequential order. Also, consider the size of each record is 128 Bytes and the size of each disk block is 512 Bytes. For the given scenario, analyze the problem of searching a record. Moreover, provide a way to solve this problem. 10

c) Implement a two-level free-space map for a file with 16 blocks (shown in figure 3(c)). Assume that 3 bits are used to store the occupancy fraction. For second level, use 1 entry for every 4 entries of the main free-space map. 10

4	2	1	4	7	3	6	5	1	2	0	1	1	0	5	6

Figure 3(a): A file with 16 blocks

4. a) Consider a static hashing system which is determined by the hash function defined as follow: $H(M) = M \ mod \ 2^k$ where $k = 3$. Construct hash index on attribute emp_no of the table 2(a). Consider bucket size is 2. 10



emp_no	.....
1688	.....
1722	.....
1741	.....
1764	.....
2444	.....
2922	.....
3292	.....
4178	.....
4919	.....
5569	.....
6036	.....

Figure 4(a): Hash structure

b) *members (id: integer, name: string, email: string, birthdate: date)* 10

reminders (id: integer, memberId: integer, message: string)

Primary key attributes are shown underlined. Create an *AFTER INSERT* trigger that inserts a reminder into the *reminders* table if the *birth date* of the member is *NULL*.

c) Briefly describe RAID level 5 and RAID level 6 with proper illustrations. 10

PART- B (Marks: 90)

(Answer any three questions including Q. No. 5)

Marks

5. a) Analyze the problems related to static hashing. Suggest some solutions to overcome these problems. 10

b) Consider the following schemas: CLO4

Customers(cust_id: integer, name: string, address: string, phone: string)

Orders(order_no: integer, cid: integer → Ref(Customers: cust_id), order_date: date, product: string, qty: integer, price: float)

Shipments(ship_id: integer, oid: integer → Ref(Orders: order_no), ship_date: date, carier_no: integer)

Primary key attributes are shown underlined. Write the SQL statements for the following:

i) Find the name of the customers along with their address. 03

ii) Find the order number of the orders which are placed between 01/12/2020 and 31/12/2020. 04

iii) List customer wise order numbers and shipment dates. 04

iv) Find the order number for those orders whose customer names begin with "a" and are at least 5 characters long. 04

c) Justify the following statement – "Limitations of file processing systems prompted the development of database system". 05

6. a) Consider the B+ tree presented in Figure 6(a) where number of pointers n = 4. 15

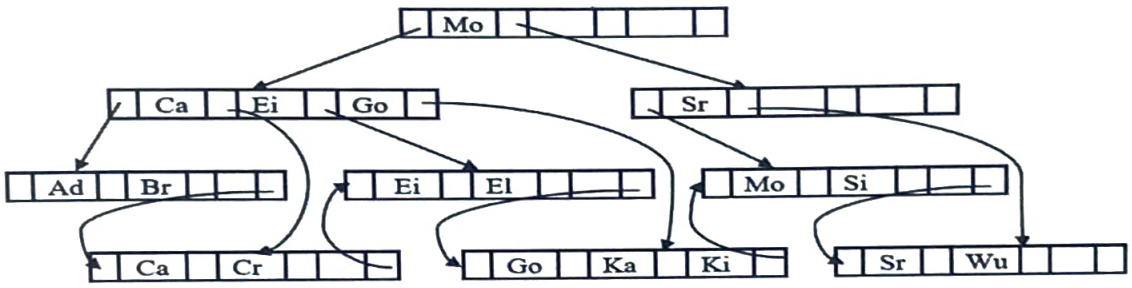


Figure 6(a): B+ tree

Draw the B+ trees after deleting each of the following search key values: Sr, Si and Go.

b) Write a stored function that takes year as input and can determine whether it is leap year or not. You may use following algorithm: 10

- If year is divisible by 4 and not divisible by 100 then it is leap year.
- Or if it is divisible by 400 then it is leap year.
- Otherwise, it is not leap year.

c) Differentiate between strong entity and weak entity. 05

7. a) Draw a wait for graph for the partial schedule shown in Figure 7(a). Is there any deadlock? If yes then, write a deadlock recovery procedure. 15

Instruction No.	T1	T2	T3
1	Lock-S(X)		
2	Read(X)		
3			Lock-X(Z)
4			Read(Z)
5	Lock-X(Z)		
6		Lock-S(Y)	
7		Read(Y)	
8		Lock-X(X)	
9			Lock-X(Y)
10	---	---	---

Figure 7(a): Given partial schedule

b) Consider the log records depicted in Table 7(b) which are written during normal operation. Write the recovery log records after system crash. 10

<T_0 Start>
<T_0, Y, 1000, 1500>
<T_0 Commit>
<T_1 Start>
<T_1, Y, 1500, 1200>
<T_2 Start>
<T_4 Start>
<T_2, Z, 400, 300>
<Checkpoint {T_1, T_4, T_2}>
<T_2, B, 600, 650>
<T_3 Start>
<T_3, X, 600, 700>
<T_1, Y, 1500>
<T_4, A, 300, 200>
<T_1 abort>

Table 7(b): Log records during normal operation

c) Briefly explain the following deadlock prevention scheme: wait-die and wound-wait. 05

8. a) Describe digital signature scheme with proper illustration. 10

b) Consider the following database table: 10

Table 8(b): Section table

Student_id	Sub_name	Teacher	Exam_name	Total_marks	Marks_obtained
10	OOP2	T1	OOP2	150	120
10	DBMS	T2	DBMS	210	160
11	OOP2	T1	OOP2	150	115

Primary key attributes are shown underlined. You are given the following constraints:

Sub_name → Teacher

Exam_name → Total_marks

Normalize the table up to 3rd normal form.

c) Explain the causes of bucket overflow. Also propose a method to handle bucket overflow. 10

Bangladesh Army University of Science and Technology (BAUST), Saidpur

Department of Computer Science and Engineering (CSE)

Final Examination, Winter 2024

Program: B.Sc. Engg. in CSE **Course Code: CSE 2205 (Syl. 2021)** **Level- 2 Term- II**

Time: 03 (Three) hours **Credit: 3.00** **Full Marks: 180**

Course Title: Database Management Systems

N.B.
- Symbols and abbreviations bear their standard meanings.
- Use separate answer script for each PART.
- Assume reasonable values for any missing data.

PART- A (Marks: 90)

(Answer any three questions including Question No. 1)

		Marks	COs/ CLOs
a)	Suppose BAUST authority has hired you to design a database for BAUST Library. Due to this reason, you have visited the organization and conducted your own analysis over existing system. Design a database based on your findings. Also reduce your design to relational schemas.	(15)	3
b)	"Indexing speeds up the search." – Provide an example to support this claim.	(10)	1
c)	Define Data Abstraction. Briefly describe different levels of Data Abstraction.	(5)	1
2. a)	Draw a wait for graph for the partial schedule shown in Table 2(a). Assess if a deadlock occurs or not. If yes then, show the recovery procedure.	(15)	

Table 2(a): Given partial schedule.

Instruction No.	T1	T2	T3
1	Lock-S(X)		
2	Read(X)		
3			Lock-X(Z)
4			Read(Z)
5	Lock-X(Z)		
6		Lock-S(Y)	
7		Read(Y)	
8		Lock-X(X)	
9			Lock-X(Y)
10	---	---	---

b)	Differentiate between clustering and non-clustering index.	(5)	
c)	Briefly explain the functional dependency in Table 2(c) and provide a solution.	(10)	

Table 2(c): Employee table.

ID	Name	Skill	Degree
101	Mr. A	OS	M.Sc.
102	Mrs. B	OS	M.Sc.
101	Mr. A	ETH	Ph.D.
103	Mr. C	HCI	M.Sc.

3. a) Consider the following schemas:

Customers(cust_id: integer, name: string, address: string, phone: string)

Orders(order_no: integer, cust_id: integer, order_date: date, product: string, qty: integer, price: float)

Shipments(ship_id: integer, order_no: integer, ship_date: date, carier_no: integer)

Primary key attributes are shown underlined. Write the relational algebraic expression for the following:

i) List order numbers along with the associated order dates. (3)

ii) Find the order numbers for orders that are shipped after May 1, 2024. (3)

iii) Find every order that shipped before May 1, 2024, along with the quantity of each order. (4)

iv) Determine every customer, their orders, and the total amount they have paid for each order. (5)

b) Differentiate between various types of attributes found in ER Model. Additionally, provide their ER diagrammatic representation. (7)

c) Define cryptography. Briefly describe the symmetric key cryptosystem with proper block diagram. (8)

4. a) Draw the precedence graph for the above schedule displayed in Table 4(a). Determine whether the schedule is conflict serializable or not. If it is not conflict serializable, then briefly explain the reason. If it is conflict serializable, then determine the serializability order using topological sort. (15)

Table 4(a): Given Schedule.

Instruction No.	T1	T2	T3
1	Read(X)		
2	X:=X-50		
3			
4			Read(Y)
5	Write(X)		Y:=X*2
6			
7			Write(Y)
8		Read(X)	
9		X:=X+30	
10	Read(Y)	Write(X)	
11	Y:=Y+10		
12	Write(Y)		
13			
14			Read(Y)
		Read(Y)	

b) Write the working principle of a disk controller. (7)

c) Describe the different types of users of the database. (8)

PART- B (Marks: 90)

(Answer any three questions including Question No. 5)

Marks COs CLO

5. a) Consider the following schemas: 4

Products(prod_sl: integer, pname: string, cat: integer)

Categories(cid: integer, cname: string)

Sale(id: integer, prod_sl: integer, cid: integer, sdate: date, qty: integer, price: float)

Primary key attributes are shown underlined. Write the SQL statements for the following:

i) Find the names of all the products. (3)

ii) Determine the total number of products sold between January 1, 2024, and May 1, 2024. Additionally, arrange the results by product serial number in ascending order. (3)

iii) List the product categories and names that contain the substring "en" at any position. (4)

iv) List all the products, along with the total amount of sales, whose total quantity exceeds the average quantity. (5)

b) Analyze the issues associated with the non-normalized database. Provide an idea on how to solve these issues. (10) 2

c) List the properties of a B+ tree. (5)

7. a) *Sales (id: integer, product: string, qty: integer, year: year)*
SalesChanges(salesId: integer,beforeQuantity: integer, afterQuantity: integer) (10)

For the above schemas, write a trigger that is automatically fired before an update event occurs for each row in the sales table. If the value in the quantity column changes, the trigger inserts a new row to record the changes in the SalesChanges table.

b) Briefly explain the purpose of applying checkpointing procedure periodically. Write the steps of performing checkpointing. (10)

c) Differentiate between primary key and foreign key. Consider the relation show in Table 8(a). Find the supper key, candidate key, and primary key. (10)

8. a) Consider the schedule depicted in Table 7(a) which consists of three transactions T1, T2 and T3. Here, the initial value for data items W, X, Y and Z was 100, 1000, 900 and 800 respectively. Suppose that the system made a checkpoint just after completing instruction number 9. A few instructions later (after instruction number 11), it failed. Now, write log records for the system. Also, write the log records which were added during recovery. (15)

Table 7(a): Given schedule.

Instruction No.	T1	T2	T3
1	Read(X)		
2	X:=X-150		
3			Read(Z)
4			Z:=Z*2
5	Write(X)		
6			Z:=Z+W
7		Read(Y)	
8		Y:=Y+300	
9	Commit		
10			Write(Z)
11		Write(Y)	
12			Commit
13		Commit	

b) Define transaction. Describe the ACID properties of a transaction. (7)

c) Describe the emergence of variable-length records in databases. Explain the mechanism of slotted page structure for handling records with varying lengths. (8)

8. a) Construct an Extendable Hash Structure using Table 8(a). Let, **Col3** represents the search key values, bucket size is 2, and hash function is the six-digit binary representation of a search key value. (20)

Table 8(a): A database table

Col1	Col2	Col3
1	M	13
2	S	60
3	W	40
4	E	38
5	G	38
6	D	49
7	K	60
8	B	60

b) Illustrate the ER diagram to relational schema reduction procedure for the followings: Weak Entity Set, Multivalued Attribute, Many to Many Relationship Set with Attribute, and Relationship Set linking a Weak Entity Set (Many) to the corresponding Strong Entity Set (One). (10)

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ıe: 03 (Three) hours **Credit: 3.00** **Full Marks: 180**

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PART- A (Marks: 90)

(Answer any three questions including Question No. 1)

	Marks	CLOs
Design a database for a bookstore. The database should store information about books, customers, suppliers, and authors. You must draw an ER diagram and reduce schemas from the diagram considering the following information: • Every author has an author's id, name, and short biography. • Every book has ISSN, title, publisher, and year, and may have one or more authors. • Every supplier can supply one or more books, and has a supplier's id, name, address, and phone number. • Every customer has a customer's id, name, address, and phone number. • A customer can purchase one or more books. • You must record a unique invoice number, quantity, unit price, and date for every purchase and supply.	(15)	3
Consider the following schemas: *employee (person id: integer, person_name: string, street: string, city: string)* *works (person id: integer, company_id: integer, salary: integer)* *company (company id: integer, company_name: string, city: string)* Primary key attributes are shown underlined. Write the **SQL statements** for the following:		4
i) Find the names and cities of all the employees.	(3)	
ii) Find companies in Dhaka.	(3)	
iii) List the names of employees who earn more than Taka 40000 and work for a company whose city ends with the substring 'pur'.	(4)	
iv) Find the names and total salaries of companies that paid more than the average salary. Arrange the output in ascending order by company name.	(5)	
Consider a relation with schema $R(A, B, C, D, E, F)$ and perform the following tasks:		
i) FD = $\{CD \rightarrow E, B \rightarrow D, E \rightarrow A, A \rightarrow BC\}$. Calculate CD^+.	(2)	
ii) FD = $\{C \rightarrow F, Q \rightarrow A, EC \rightarrow D, A \rightarrow B\}$. Find all candidate keys.	(4)	
iii) FD = $\{CD \rightarrow A, C \rightarrow D, E \rightarrow A, B \rightarrow CD, EC \rightarrow B\}$. Find all candidate keys.	(4)	
iv) FD = $\{A \rightarrow B, B \rightarrow F, C \rightarrow D, D \rightarrow E\}$. Determine whether the given relation is in 3NF or not. If not, then convert it into 3NF.	(5)	
Differentiate between UNION and UNION ALL with proper example.	(5)	
List the steps of database design process. Write the necessary conditions for the multivalued dependency. Provide an example.	(10)	

3. a) Consider the following schemas:

Supplier(sid: integer, sname: string, address: string)

Part(pid: integer, pname: string, colour: string)

Catalog(sid: integer, pid: integer, cost: real)

Primary key attributes are shown underlined. Write the **relational algebraic expression** for the following:

i) List the names of the red color parts. (3)

ii) Find the IDs of suppliers who supply some red or green color part. (3)

iii) Find the total cost for each part. (4)

iv) Find the IDs of suppliers who supply only red-colored parts. (5)

b) Differentiate between various types of attributes found in ER Model. Additionally, provide their ER diagrammatic representation. (7)

c) Define DBMS. "Three-tier applications are more appropriate for large applications, and for applications that run on the World Wide Web." – Justify the statement. (8)

4. a) *WorkCenters(id, name, capacity)* (10)
WorkCenterStats(totalCapacity)
For the schemas above, write a trigger that can update the total capacity of the WorkCenterStats table before a new work center is inserted into the WorkCenters table.

b) Differentiate between primary key and foreign key. Identify with appropriate justifications, supper keys, candidate keys, composite keys, and primary key from Table 4(b). (10)

Table 4(b): Registration table.

ID	CourseCode	Semester	Year
101	CSE 2201	Summer	2024
101	CSE 2205	Winter	2024
102	CSE 2201	Summer	2024
102	CSE 2205	Winter	2024
103	CSE 2207	Winter	2024

c) List the purposes of normalization. Briefly explain different types of anomalies that occur in an unnormalized database with examples. (10)

PART- B (Marks: 90)

(Answer any three questions including Question No. 5)

Mark

5. a) Describe RAID briefly with the proper explanation of the advantage of using RAID technology in database system. (8)

b) Imagine a healthcare system with an Electronic Health Records (EHR) database. The database stores patient information, medical histories, and other sensitive data. This data is crucial for doctors and healthcare professionals to access at any moment, especially in emergencies. (14)
In this situation you need to choose a suitable recovery system from different RAID levels.

i) Identify the RAID level that you will choose.

ii) Describe the reasons against your choice.

iii) Describe the chosen RAID level briefly with necessary technical diagram(s).

c) Write short notes on below topics: (8)

i) Big Data

ii) Cloud Computing

) Draw and explain the basic block-diagram of application-database-storage communication process including other necessary components to do different transaction on a relational database. (12)

) Briefly explain three different types of database files. Explain with necessary diagram—"Explore the data rows are stored in different data pages." (10)

) Write short notes on below topics: (8)

i) Clustered Index

ii) Data Page

a) Write major advantages of using indexing in database management system. (8)

b) Describe the B-Tree structure. Explain B-Tree structure is used in database record storing and organization. (8)

c) Perform delete operation on the below B-Tree (Figure 7(c)) for values 64, 23, 20 serially and draw the updated tree after deletion. (14)

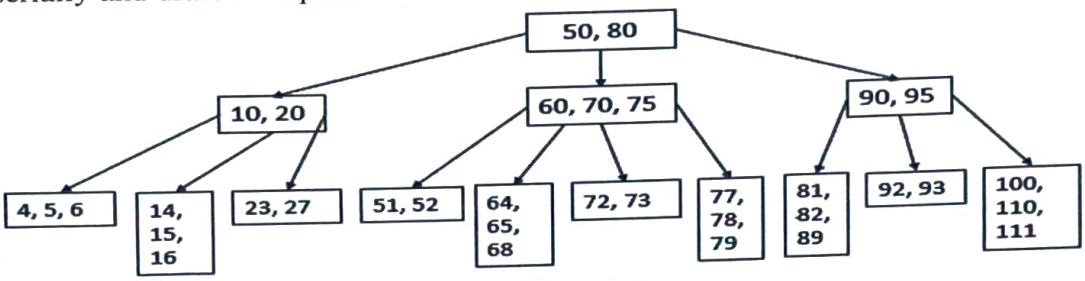


Figure 7(c): B-Tree

a) Justify the following statements from technical point of view where all statements describe the query performance optimization techniques. (12)

i) Avoid SELECT * and retrieve only necessary columns

ii) Utilize stored procedures and functions

iii) Use UNION ALL instead of UNION

b) Give a brief explanation of ACID properties. (6)

c) Carefully look at the following tables. (12)

- **Products** (ID INT PRIMARY KEY, Name VARCHAR(50) NOT NULL, Category VARCHAR(50), SubCategory VARCHAR(50))
- **Area** (ID INT PRIMARY KEY, Name VARCHAR(50) NOT NULL, District VARCHAR(50), Thana VARCHAR(50))
- **Customer** (ID INT PRIMARY KEY, Name VARCHAR(50) NOT NULL, Gender VARCHAR(50), City VARCHAR(50))
- **Time** (ID INT PRIMARY KEY, Day VARCHAR(50), Month VARCHAR(50), Quarter VARCHAR(50), Year VARCHAR(50))

In a retail business, analyzing sales data across multiple dimensions (e.g., time, location, and product category) is crucial for decision-making. Multidimensional databases (MDDBs) allow quick aggregation and summary operations, such as total sales by region, monthly growth, or category-wise sales breakdowns.

To support the above scenario conventional database schema doesn't perform better. In that case, identify the best filled schema. Draw the diagram of your suggested database schema.

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Time: 03 (Three) hours **Credit: 3.00** **Full Marks: 180**

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PART- A (Marks: 90)

(Answer any three questions including Question No. 1)

		Marks	COs/CLOs
a)	Design an Entity-Relationship (ER) model for a school management system. The system should include entities for Student, Teacher, Course, Classroom, and Department. You must draw an ER diagram and reduce schemas from the diagram considering the following information: • Entities must be strong type. • Include necessary attributes for each entity. • Relationships are binary type and may have attributes if necessary. • Include necessary cardinalities and any other constraints. • Use standard notations for drawing the ER diagram.	(15)	3
b)	Consider the following schemas: *employee (person id: integer, person_name: string, street: string, city: string)* *works (person id: integer, company_id: integer, salary: integer)* *company (company id: integer, company_name: string, city: string)* Primary key attributes are shown underlined. Write the **SQL statements** for the following:		4
	i) Find the names and cities of all the companies.	(03)	
	ii) Find employee names that contains 'al' in any position.	(03)	
	iii) List the top five salaried employees and their cities. Arrange the output in ascending order by salary.	(04)	
	iv) Find the names and total salaries of companies that paid more than the average salary.	(05)	
a)	Consider a relation with schema $R(A, B, C, D, E, F)$ and perform the following tasks:		
	i) FD = $\{AB \rightarrow C, BC \rightarrow AD, D \rightarrow E, CF \rightarrow B\}$. Prove that $AB \rightarrow F$ holds.	(02)	
	ii) FD = $\{AB \rightarrow DE, E \rightarrow C\}$. Find all candidate keys.	(04)	
	iii) FD = $\{C \rightarrow F, E \rightarrow A, A \rightarrow B, EC \rightarrow D\}$. Find all candidate keys.	(04)	
	iv) FD = $\{A \rightarrow B, B \rightarrow F, C \rightarrow D, D \rightarrow E\}$. Determine whether the given relation is in 3NF or not. If not, then convert it into 3NF.	(05)	
b)	Define Database, Query, and Schema. Differentiate between DDL, DML, and DCL.	(10)	
c)	Consider the following relations and perform **R1/R2**. Justify your answer.	(05)	

Table 2(c)-I: Relation R1

Col1	Col2	Col3
A1	B2	C1
A4	B5	C2
A1	B2	C2
A4	B5	C4

Table 2(c)-II: Relation R2

Col1	Col2
A1	B2
A4	B5

3. a) Consider the following Relations:

Table 3(a)-I: Products.

Prod_id	Name	Cat_id
1	Mouse	2
2	Laptop	1
3	Keyboard	2
4	Stand	3
5	Mobile	1

Table 3(a)-II: Categories.

Cat_id	Cat_name
1	Electronics
2	Accessories
3	Office Supplies
4	Stationary

Table 3(a)-III: Orders.

Order_id	Prod_id	Qty	Price
1	2	2	180
2	5	10	200
3	3	20	50
4	1	50	60
5	4	50	30

Primary key attributes are shown underlined. Demonstrate the output of the following **relational algebraic expression**:

i) $\pi_{Name}(\sigma_{Cat_id=1}(Products))$ (03)

ii) $\pi_{Name,Cat_name,Qty,Price}(\sigma_{Qty>20}(Orders) \bowtie Products \bowtie Categories)$ (03)

iii) $\pi_{Cat_name}(Categories) - \pi_{Cat_name}(\sigma_{Prod_id>2}(Products \bowtie Categories))$ (04)

iv) $_{Cat_name}\mathcal{G}_{SUM(Qty \times Price)}(Orders \bowtie Products \bowtie Categories)$ (05)

b) Briefly explain the usages of subquery with proper example. State the meaning of the following patterns: 'T_m', '%_20%', and '%_on'. (10)

c) List the steps of drawing an ER diagram. (05)

4. a) *Customers(CustNo, Name, CreditLimit)*
Write a stored procedure named *GetCustomerLevel()* based on the above schema. It should accept two parameters: *CustNo (IN)* and *CustomerLevel (OUT)*, and demonstrate customer level (Silver, Gold, and Platinum) as output. You must implement the following logic: (10)

- If the credit limit is greater than 50000, then the customer level is Platinum.
- If the credit limit is between 10000 and 50000, then the customer level is Gold.
- And if the credit limit is below 10000, then the customer level is Silver.

b) Differentiate between primary key and foreign key. Identify supper keys, candidate keys, composite keys, and primary key from Table 4(b). (10)

Table 4(b): Registration table.

ID	CourseCode	Semester	Year
101	CSE 2201	Summer	2024
101	CSE 2205	Winter	2024
102	CSE 2201	Summer	2024
102	CSE 2205	Winter	2024
103	CSE 2207	Winter	2024

c) Briefly explain the following Mapping cardinalities with their ER diagrammatic representations: One-to-One, One-to-Many, Many-to-One, and Many-to-Many. (10)

PART- B (Marks: 90)

(Answer any three questions including Question No. 5)

		Marks	CLOs
a)	Describe the concept of locking in database transactions. Compare and contrast the Basic, and Rigorous Two-Phase Locking (2PL) protocols in terms of their behavior and impact on concurrency and recoverability.	(08)	1
b)	Analyze Schedule 1 and Schedule 2 as presented in **Table 5**. Considering the transaction commit order, determine which schedule is recoverable with proper reason. Additionally, evaluate which schedule is more efficient in terms of CPU time and justify your answer.	(15)	2

Table 5: (a) Schedule 1 and (b) Schedule 2.

T1	T2
Read(A)	
A: =A-50;	
Write (A)	
	Read(A)
	A=A+500
	Write(A)
Failure	
commit;	
	commit;

(a) Schedule 1

T1	T2
Read(A)	
A: =A-50;	
Write (A)	
commit;	
	Read(A)
	A=A+500
	Write(A)
Failure	
	commit;

(b) Schedule 2

c)	Refer to **Table 5** (c), User Demographics and Subscription Types, to answer the following questions related to bitmap indexing.	(07)	2

Table 5: (c) User Demographics and Subscription Types.

ID	AgeGroup	Subscription
1	Young	Premium
2	Adult	Free
3	Senior	Basic
4	Adult	Premium
5	Young	Free

i) Create a bitmap index for AgeGroup and Subscription.
ii) Write bitwise operations to find all Young users with a Free subscription.

You are given a dataset containing 2000 student records. Each record consists of the following attributes:

- **SID** (Student ID): 10 bytes
- **Name**: 100 bytes
- **Department**: 18 bytes

Additional information: Each data block is 512 bytes in size. Each record is stored without any pointers. In the index structure, each index entry contains a search key (SID) and a 6-byte pointer to a data block.

a)	Determine the time complexity (in terms of block accesses) for searching a specific record when— the records are stored in an unordered manner and no indexing is applied.	(05)	
b)	Determine the time complexity for searching a specific record sorted by **SID**, assuming no indexing is applied.	(10)	
c)	Assuming an appropriate indexing technique is used to minimize block access, identify the levels of index are required to reduce the number of blocks accessed during a search? In that case, determine the number the number of blocks need to be accessed in total to perform a search?	(15)	
a)	Construct a B+ Tree by inserting the following sequence of keys in the given order: 25, 27, 32, 2, 7, 35, 8, 23, 17, 26, 13, 9, 6. Assume that the maximum degree (order) of the B+ Tree is 3.	(15)	

b) Draw a wait-for graph for the partial schedule shown in **Table 7**. The Tx column indicates the transactions involved in the schedule. Analyze whether a deadlock occurs. If a deadlock is detected, suggest an appropriate recovery technique. Assume that A, B, C, and D represent data items.

Table 7: Partial Schedule 3.

Tx									
T1	S(A)	S(D)		S(B)					
T2			X(B)				X(C)		
T3					S(D)	S(C)			X(A)
T4								X(B)	

c) Explain the different types of databases by highlighting their key features, data models, and typical areas of application. Provide suitable examples to support your explanation.

8. a) Analyze the given transaction schedule by constructing its precedence graph. Determine whether the schedule is conflict serializable. If it is, identify a conflict-equivalent serial schedule by performing a topological sort on the graph.

Table 8: Partial Schedule 4.

T1	T2	T3
Read(A)		
		Read(B)
		Read(A)
	Read(B)	
	Read(C)	
		Write(B)
	Write(C)	
Read(C)		
Write(A)		
Write(C)		

b) Apply the extendible hashing technique to insert the following elements into a hash structure: 16, 4, 6, 22, 24, 10, 31, 7, 9, 20, 26. Assume the bucket size is 3. Clearly show the structure after each insertion and any changes to the directory or global/local depth.

c) Write concise explanatory notes on the following topics. Provide relevant examples where appropriate to enhance clarity and understanding:

i) Spatial Indexing
ii) Temporal Indexing
iii) Cryptography

Bangladesh Army University of Science and Technology (BAUST), Saidpur

Department of Computer Science and Engineering (CSE)

Final Examination, Winter/ Summer 2025

Program: B.Sc. Engg. in CSE **Course Code: CSE 2205 (Syl. 2021)** **Level- 2** **Term- II**

Time: 03 (Three) hours **Credit: 3.00** **Full Marks: 180**

Course Title: Database Management Systems

N.B
- Symbols and abbreviations bear their standard meanings.
- Use a separate answer script for each Part
- Assume reasonable values for any missing data.

PART- A (Marks: 90)

(Answer any three questions including Question No. 1)

		Marks	COs/ CLOs
a)	Design an Entity–Relationship (ER) model for a Hotel Management System. The system should include entities for Guest, Room, and Reservation. Draw an ER diagram and reduce schemas from the diagram, considering the following information: • All entities must be strong entities. • Include necessary attributes for each entity. • The Guest entity must include a mandatory Address attribute and a Contact Number attribute, where a guest can have multiple contact numbers. • Relationships must be binary and may include attributes if necessary. • Include appropriate cardinalities and any required constraints. • Use standard ER notation when drawing the diagram.	(15)	3
b)	Consider the following relation schema: `R(StudentID, StudentName, CourseID, CourseName, InstructorID, InstructorName, Department, RoomNo)` The set of functional dependencies (FDs) are as follows: • `StudentID → StudentName` • `CourseID → CourseName, Department` • `InstructorID → InstructorName, RoomNo` • `CourseID → InstructorID` Answer the following Questions: i) Analyze the given functional dependencies and determine whether the relation R satisfies the conditions for 3NF. ii) If the relation is not already in 3NF, perform a 3NF decomposition and provide the resulting normalized relational schemas.	(15)	2

a)	Consider the following database schemas:
	`Doctors(DoctorID, DoctorName, Specialization)`
	`Appointments(AppointID, PatientID, DoctorID, AppointDate, Fee)`
	Create a stored procedure named **`GetDoctorEarning`**`(p_DoctorID)` that returns summary information for a doctor. The procedure must:
	• Count how many patients the doctor has visited
	• Calculate the total amount of fees earned by the doctor
	• Identify the earliest appointment date
	• Identify the most recent appointment date
		(15)	

b)	Differentiate between a function and a procedure. Highlight their key differences with examples.	(10)	
c)	Explain the problems that may arise in a database without normalization. Justify why normalization is necessary.	(05)	

3. a) Consider the following database schemas for a company:

```
Employees(EmployeeID, EmployeeName, DepartmentID, ManagerID)
Departments(DepartmentID, DepartmentName, Location)
Projects(ProjectID, ProjectName, DepartmentID)
```

i) Write an SQL query to display the EmployeeName, DepartmentName, and ProjectName for all employees and their assigned projects. Include employees who may not be assigned to any project.

ii) Write a query to list all projects, including those that do not have any employees assigned.

iii) Write a query to find the names of employees whose names contain the letters "son" anywhere.

b) A bank wants to design a new online banking system. Compare how this system would differ if implemented using a 2-tier and a 3-tier architecture. Justify which architecture is more suitable for scalability, security, and maintainability.

c) Define Primary Key, Alternative Key, Super Key, Foreign Key, and Candidate Key, with appropriate examples.

4. a) Consider a database for a library with the following schemas:

```
Books(BookID, Title, Author, QuantityAvailable)
Transactions(TransactionID, BookID, MemberID, IssueDate,
ReturnDate)
```

i) Write an AFTER INSERT trigger on the Transactions table that automatically decrements QuantityAvailable in the Books table whenever a new book is issued (i.e., a row is inserted with a non-null IssueDate).

ii) Write an AFTER UPDATE trigger on the Transactions table that automatically increments QuantityAvailable whenever a book is returned (i.e., when ReturnDate is updated).

b) Consider a database with the tables:

```
Students(StudentID, StudentName, CourseID, Mark)
Courses(CourseID, CourseName)
```

Create a view that displays the *StudentName*, *CourseName*, and *Mark* for all students enrolled in courses. Ensure the view only includes students with a mark of 80 or higher.

c) Without integrity constraints a database may become inconsistent or unreliable. Justify your answer with examples.

PART- B (Marks: 90)

(Answer any three questions including Question No. 5)

5. a) Apply mathematical and algorithmic concepts to construct a B+ Tree of order 4 by inserting the keys in the following sequence: 2, 50, 60, 70, 45, 66, 1, 65, 5, 3, and determine the minimum number of keys that can be stored in an internal node and a leaf node.

b) Consider the following table structure for a spatial database in PostgreSQL:

```
Places(
    PlaceID   SERIAL PRIMARY KEY,
    Name      VARCHAR(100),
    geom      GEOMETRY(Point, 4326)
)
```

Using this table, implement and manage a spatial database solution in PostGIS by completing the tasks below:

i) Enable spatial support in PostgreSQL.

ii) Create the Places table as shown above.

iii) Insert at least three sample records, including one named 'BAUST'.

iv) Write an SQL query to find all locations within a chosen distance of 500 meters from 'BAUST'.

c) Differentiate Between Hash-Based and B+ Tree Indexing (05)

a) Consider the following schedule given on Table 6(a) of two transactions, T1 and T2: (15)

Table 6(a): Schedule 1

T1	T2
Read(A)	
Write(A)	
	Read(A)
	Write(A)
Read(B)	
Write(B)	
	Read(B)
	Write(B)

Apply conflict-serializability principles to rearrange the schedule by swapping non-conflicting operations, and determine whether these swaps can produce a conflict-equivalent serial schedule or if the schedule is not conflict-serializable.

) Explain the different stages of a database transaction. Using an example, illustrate how a transaction can be in an inconsistent state during its execution. (15)

) A company is developing a new employee management system and wants to implement secure authentication for its employees. The system will store sensitive information, including salaries, bank account details, and personal records. The company wants to ensure that data remains confidential, secure, and accessible only to authorized users. (20)
Based on this scenario, answer the following:

i) Explain the difference between hashing and encryption.
ii) Discuss which method is suitable for:
   a) Authentication (e.g., storing passwords)
   b) Sensitive data storage (e.g., salary, bank account information)
iii) Illustrate how the sensitive data can be stored and retrieved securely, including the necessary SQL statements.
iv) Explain how the system can ensure secure authentication for users.

) Explain the sequence of steps performed by the ARIES recovery algorithm to restore the database after a crash. (10)

) Consider the following transaction log on Table 8(a) and answer the given questions. (20)

Table 8(a): Transaction Log of Transactions T1 and T2

Log Record
<T1 start>
<T1, X, 500, 450>
<T2 start>
<T2, Y, 300, 250>
<T1, Z, 800, 900>
<T1 commit>
<T2, W, 1000, 950>
CRASH

i) Identify which transactions need REDO and which need UNDO after recovery.
ii) Show the recovery actions step-by-step.
iii) Write the final database values of X, Y, Z, and W after recovery.
iv) Write the new log entries that would be added during the recovery process

b) Explain the purpose and usage of checkpoints in transaction recovery. Illustrate yo answer with an appropriate example that demonstrates how checkpoints are utilized in t log to recover a database after a failure.